



WORLD UNIVERSITY OF BANGLADESH

DISCIPLINE INNOVATION KNOWLEDGE

DEPARTMENT OF AUTOMOBILE ENGINEERING

Experiment No : 03

Experiment Name : Verification of De Morgans Theorem,

Course Name : Digital Electronic Lab

Course Code : 07142106

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Date of Performance

Date of Submission

Experiment No: 03

Experiment Name: Verification of De Morgans Theorem.

Objective:

To experimentally verify De Morgans Theorem using logic gates and compare the truth tables of the equivalent Boolean expression.

Theory:

De Morgan Theorem provides a powerful way to relate the three basic logic operation (AND, OR and NOT)

There are two primary laws:

First law:

The complement of a product of variables is equal to the sum of the complement of the variables.

$$\overline{A \cdot B} = \overline{A} + \overline{B}$$

(A NAND gate is equivalent to a Bubbled-OR gate)

Second Law:

The complement of a sum of variables is equal to the product of the complement of the variable.

$$\overline{A+B} = \overline{A} \cdot \overline{B}$$

(A NOR gate is equivalent to a Bubbled-AND gate)

Apparatus:

1. Logic Trainer Kit
2. DC power supply
3. IC 7404 (NOR gate)
4. IC 7408 (AND gate)
5. IC 7432 (OR gate)
6. IC 7400 (NAND gate)
7. IC 7402 (NOR gate)
8. Connecting wires and LEDs.

Procedure:

Setup: Place the required ICs on the breadboard and connect pin 14 to V_{CC} (5V) and pin 7

Verify First Law:

1. Construct the circuit for the left hand side (LHS): $\overline{A \cdot B}$ using a NAND gate.
2. Construct the circuit for the right hand side (RHS): $\overline{A} + \overline{B}$ using NOT gates and OR gates.
3. Apply input combination (00, 01, 10, 11) and record the output of both.

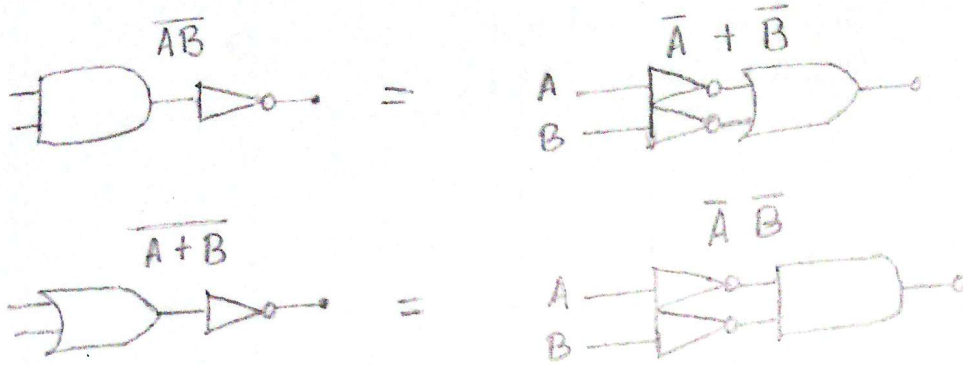
Verify Second Law:

1. Construct the LHS: $\overline{A + B}$ using a NOR gate.
2. Construct the RHS: $\overline{A} \cdot \overline{B}$ using NOT gate and an AND gate.
3. Apply input combination and record the output.

Observation:

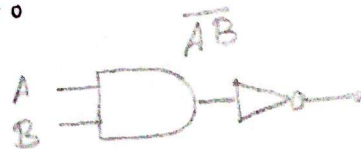
combination the output columns for LHS or RHS in both cases.

Logic Diagram:

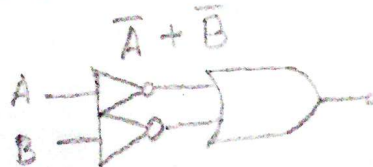


First Condition:

Left Hand Side:

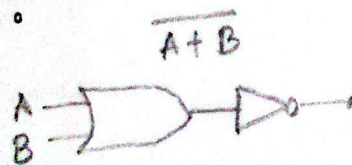


Right Hand Side:

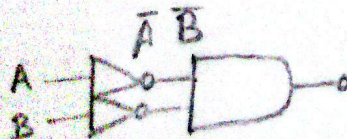


Second Condition:

Left Hand Side:



Right Hand Side:



Truth Table for the first condition:

A	B	$Y = A \cdot B$ (LHS)	$Y = \overline{A} + \overline{B}$ (RHS)
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0

Truth table for the second condition:

A	B	(LHS) $Y = A + B$	(RHS) $Y = \overline{A} \cdot \overline{B}$
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

Result: Since the columns for $\overline{A} + \overline{B}$ and $\overline{A} \cdot \overline{B}$ are identical the second condition is verified.

Final Result Summary:

Table-1: Shows that a NAND gate is logically equivalent to an OR gate with inverted inputs.

Table-2: Shows that a NOR gate is logically equivalent to an AND gate with inverted inputs.

Discussion:

The experimental result should align perfectly with the theoretical truth table. During the experiment ensure that the ICs are handled carefully as they are sensitive to static and incorrect voltage polarity.

Conclusion:

The De Morgans Theorem were successfully verified. The output of the LHS circuit matched the RHS circuit for all input combination. Confirming that the mathematical identities hold true in physical logic hardware.